CYBER SECURITY INTERNSHIP

DATE: 03-09-2026

TASK 4

SETUP AND USE A FIREWALL ON WINDOWS/LINUX

TOOLS USED:

WINDOWS DEFENDER FIREWALL

WINDOWS POWERSHELL

COMMAND PROMPT

NMAP

1. INTRODUCTION

A Firewall is a network security system, available as hardware or software, that monitors and controls incoming and outgoing traffic based on predefined rules. It acts like a security guard, filtering data packets to either:

Accept: Allow the traffic.
Reject: Block with an error response.
Drop: Block silently without response.

For this task, **Windows Defender Firewall** was used to create and test an inbound rule that blocks TCP port 23, which is commonly associated with the Telnet protocol.

2. TOOLS USED

Tool	Purpose
Windows Defender Firewall	To manage and configure inbound and outbound firewall rules.
Windows PowerShell	To create, verify and remove the firewall rule.
Command Prompt	To find the local IPv4 address.
Nmap	To test the status of TCP port 23 on the local system.

3. WORKING OF FIREWALL

A firewall monitors and filters incoming and outgoing network traffic based on predefined security rules. It decides whether a connection should be allowed or blocked.

a) **Traffic enters or leaves the computer**: All network connections to and from the system are checked by the firewall.

b) **Traffic is inspected:** The firewall examines the traffic based on rules such as IP address, port number, protocol and traffic direction.

c) **Rules are applied:** If the traffic matches an allow rule, it is permitted. If it matches a block rule, it is denied.

d) **Traffic is logged:** Firewall activity, including blocked or allowed connections, can be recorded in firewall logs for monitoring and troubleshooting.

e) **Default behaviour is applied:** If no specific rule matches the traffic, the firewall follows its configured default policy. A restrictive default policy helps prevent unauthorized network access.

4. EXAMPLE FROM THIS TASK:

- An inbound rule was created to block **TCP port 23 (Telnet)**.
- The firewall rule was verified using Windows PowerShell to confirm that it was enabled, applied to inbound TCP traffic and targeted port 23.
- The blocked port was tested using Nmap. The scan showed 23/tcp closed telnet, meaning the computer was reachable but no Telnet service was listening on that port.
- After completing the test, the temporary firewall rule was removed and its removal was verified, restoring the previous firewall configuration.
- This demonstrates how a firewall can restrict incoming traffic to a specific network port and service.

5. CONCLUSION

This task helped me understand the **basic working** and **configuration of Windows Firewall**. I learned how to create, verify, test and remove a firewall rule for a specific port. It also helped me understand how firewalls control network traffic and how tools like **PowerShell** and **Nmap** can be used to manage and test firewall configurations.